ModelIt!

Student Activity Pack

Earth Has a Fever

Why 2 Degrees Could Change Everything

Grade 7 | NGSS MS-ESS3-5 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think causes Earth's temperature to change?

2 Why do you think ice at the North and South Poles is melting?

3 What is the greenhouse effect? Draw what you think it looks like.

4 How do human activities affect Earth's temperature?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

CO2 Concentration | Solar Energy Input | Average Global Temperature | Ice Sheet Coverage

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Pre-Industrial

CO_2 at low historical levels, stable solar input

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Current Trajectory

CO_2 locked at high and rising levels

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Emission Reduction

CO_2 reduced to moderate levels

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Methane Emissions — Methane is a greenhouse gas 80x more potent than CO_2, released by livestock, landfills, and permafrost

Ocean Absorption — Oceans absorb about 30% of CO_2 from the atmosphere, but this makes them more acidic

Albedo Effect — The reflectivity of Earth's surface — ice reflects sunlight while dark ocean absorbs it

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If Earth's temperature rises by just 2 degrees Celsius, why could it change everything? Use evidence from your simulation to support your answers.

1 Why is a 2°C rise in global temperature considered dangerous when daily temperatures vary by much more?

2 How does the melting of ice sheets create a positive feedback loop that accelerates warming?

3 What evidence from the last 100 years shows that human activity is driving climate change?

4 How does your model help you understand why some people say climate change is 'accelerating'?

5 What would it take to reverse the trend your model shows?

8 STEM Challenge: Design a School Climate Action Plan

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YOUR ENGINEERING MISSION

Create an evidence-based climate action plan for your school that reduces carbon emissions and educates the community, using data from your model.

THE SCENARIO

Your school district wants to reduce its carbon footprint by 50% in 10 years. The superintendent needs your team to propose specific actions with evidence showing they'll work.

Engineering Design Thinking

1. WHAT ARE THE BIGGEST SOURCES OF CO2 EMISSIONS AT YOUR SCHOOL?
2. WHICH REDUCTION STRATEGIES ARE MOST PRACTICAL AND COST-EFFECTIVE?
3. HOW WILL YOU EDUCATE OTHER STUDENTS ABOUT CLIMATE SCIENCE?
4. HOW WILL YOU MEASURE WHETHER YOUR PLAN IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Climate Scientists analyze climate data and build complex models to predict future conditions. They work for NOAA, NASA, universities, and environmental organizations, earning $80,000–$140,000/year.